

Alejandro Garcia Baltazar

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Summary

CS graduate actively transitioning into firmware and embedded systems. Hands-on experience with baremetal STM32 and ESP32 development, MOSFET switching, and ESP-IDF. Background in technical support and full-stack development. Open to junior firmware and embedded roles.

Education

Bachelor of Science, Computer Science – Aug 2024 | Grantham University

- Relevant coursework: Data Structures and Algorithms, Software Engineering, Computer Networks, Security Operations, Programming in JavaScript, Advanced Programming in C++, Mobile Application Development, Operating Systems, System Analysis and Design, Project Management.

Experience

unWired Broadband, Level 1 Technical Support

January 2023 – Present

- Monitored and analyzed network traffic data to troubleshoot connectivity issues and optimize network performance for over 100+ customers weekly.
- Documented technical procedures and solutions in internal knowledge bases for future reference, improving overall team efficiency and customer satisfaction.
- Developed familiarity with networking hardware, signal troubleshooting, and low-level connectivity diagnostics relevant to IoT systems.

Projects

ESP32 SELF-WATERING PLANTER | ESP-IDF | [GITHUB](#)

- **Description:** WiFi-connected pump controller using ESP-IDF on ESP32.
 - Hosts an HTTP web server for browser-based pump control.
 - GPIO configured via ESP-IDF driver APIs.
 - Designed and soldered a custom circuit board with 3D printed enclosure for the final build.

Tech Used: C, ESP-IDF, FreeRTOS, ESP32, PlatformIO

STM32 SELF-WATERING PLANTER | BAREMETAL C | [GITHUB](#)

- **Description:** Configured GPIO, RCC, and MOSFET switching directly via STM32 registers without HAL.
- Implemented pull-down networks and gate resistors for safe inductive load control.
- Learned firsthand why flyback diodes are non-negotiable on inductive loads — the hard way.

Tools used: Baremetal C, ARM Cortex-M0, STM32CubeIDE, MOSFET Drive

Writing / Community

TECHNICAL BLOG — [MEDIUM](#) (AGEE.BALTAZAR)

Documenting embedded systems projects, debugging processes, and lessons learned while building hands-on STM32/ESP32 experience. Topics include baremetal GPIO, MOSFET switching, and peripheral wiring guides for beginners.

Skills

Technical Skills

- Baremetal C, ARM Cortex-M, STM32, ESP32, ESP-IDF, FreeRTOS, GPIO/RCC/SPI/I2C, MOSFET switching, PlatformIO, STM32CubeIDE, VSCode